

EXP NO: 1

8-BIT ADDITION

OUTPUT :

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers Flag Load me at

Register	Value	Flag
A	03	S 0
BC	01 00	Z 0
DE	00 00	AC 0
HL	00 00	P 1
PSW	00 00	C 0
PC	42 0C	
SP	FF FF	
Int-Reg	00	

Decimal - Hex Conversion

Decimal: 0 Hex: 0

To Hex To Dec

I/O Ports

0 - + 00

Update Port Value

Memory

8052 - + 02

Update Memory

1 LDA 8050
2 MOV B,A
3 LDA 8052
4 ADD B
5 STA 8054
6 HLT

Start 8050 OK

Address (Hex)	Address	Data
1F72	8050	1
1F73	8051	0
1F74	8052	2
1F75	8053	0
1F76	8054	3
1F77	8055	0
1F78	8056	0
1F79	8057	0
1F7A	8058	0
1F7B	8059	0
1F7C	8060	0
1F7D	8061	0
1F7E	8062	0
1F7F	8063	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle

EXP NO: 2

8-BIT SUBTRACTION

OUTPUT :

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers

Register	Value
A	01
BC	04 00
DE	00 00
HL	00 00
PSW	00 00
PC	42 0C
SP	FF FF
Int-Reg	00

Flag

Flag	Value
S	0
Z	0
AC	0
P	0
C	0

Load me at

```
1 LDA 8000
2 MOV B,A
3 LDA 8001
4 SUB B
5 STA 8002
6 RST 1
```

Decimal - Hex Conversion

Decimal: 0 Hex: 0

To Hex To Dec

I/O Ports

0 - + 00

Update Port Value

Memory

8001 - + 05

Update Memory

Start 8000 OK

Address (Hex)	Address	Data
1F40	8000	4
1F41	8001	5
1F42	8002	1
1F43	8003	0
1F44	8004	0
1F45	8005	0
1F46	8006	0
1F47	8007	0
1F48	8008	0
1F49	8009	0
1F4A	8010	0
1F4B	8011	0
1F4C	8012	0
1F4D	8013	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle

EXP NO: 3

8-BIT MULTIPLICATION

OUTPUT :

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers

Register	Value
A	02
BC	04 00
DE	00 04
HL	00 08
PSW	00 00
PC	42 16
SP	FF FF
Int-Reg	00

Flag

Flag	Value
S	0
Z	1
AC	0
P	1
C	0

Load me at

1 LDA 2200
2 MOV B,A
3 MVI D,00
4 LDA 2201
5 MOV C,A
6 LXI H,0000
7 BACK: DAD D
8 DCR C
9 JNZ BACK
10 SHLD 2202
11 HLT

Memory

Address (Hex)	Address	Data
0898	2200	4
0899	2201	2
089A	2202	8
089B	2203	0
089C	2204	0
089D	2205	0
089E	2206	0
089F	2207	0
08A0	2208	0
08A1	2209	0
08A2	2210	0
08A3	2211	0
08A4	2212	0
08A5	2213	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle

02:35 01-08-2026

EXP NO: 5

16-BIT ADDITION

OUTPUT :

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers

Register	Value
A	0A
BC	05 00
DE	00 00
HL	00 00
PSW	00 00
PC	42 17
SP	FF FF
Int-Reg	00

Flag

Flag	Value
S	0
Z	0
AC	0
P	1
C	0

Load me at

1 LDA 3050
2 MOV B,A
3 LDA 3051
4 ADD B
5 STA 3052
6 LDA 3053
7 MOV B,A
8 LDA 3054
9 ADC B
10 STA 3055
11 HLT

Decimal - Hex Conversion

Decimal	Hex
0	0

To Hex To Dec

I/O Ports

Port	Value
0	00

Update Port Value

Memory

Address	Value
3054	05

Update Memory

Start 3050 OK

Address (Hex)	Address	Data
0BEA	3050	2
0BEB	3051	3
0BEC	3052	5
0BED	3053	5
0BEE	3054	5
0BEF	3055	10
0BF0	3056	0
0BF1	3057	0
0BF2	3058	0
0BF3	3059	0
0BF4	3060	0
0BF5	3061	0
0BF6	3062	0
0BF7	3063	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle